You need to know the following topics for test 2 of Signals and Systems

1- Find the Laplace transform $X(s)$ for signal $x(t)$ using the integral.

2- Find the Laplace transform $X(s)$ for signal $x(t)$ using the Laplace properties and table.

3- Find the signal $x(t)$ by the inverse Laplace transform of $X(s)$ using the partial fraction expansion, Laplace properties, and the table.

4- Given the differential equation of a system, the input $x(t)$, and the initial conditions of the system, you should be able to find the transfer function $H(s)$, the zero-input response, the zero-state response, the forced response, the natural response, or the total response for a given input $x(t)$.

5- Given the system differential equation or the transfer function $H(s)$, you should be able to find the frequency response $H(j\omega)$, its magnitude and phase. If a dc or sinusoidal input is given then you should be able to find the output $y(t)$ and the steady state response.

6- Given the system differential equation or the transfer function $H(s)$, you should be able to find the roots and zeros of the system and determine its external and internal stability.

7- Given a transfer function $H(s)$, you should be able to draw the canonical Form II realization.

8- Given $H(s)$, you should be able to plot the poles and zeros and sketch of the magnitude of $H(\omega)$.